

AIN SHAMS UNIVERSITY-
FACULTY OF ENGINEERING



[MCT 131]: Introduction to Mechatronics

[Report]

Project [2]: Batmobile robot

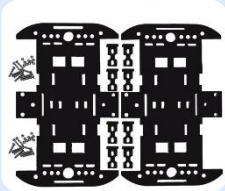
No.	Name	ID
1.	Abdullah Mahmoud Abdel Kaream	2100658
2.	Musa Gamal El-Shenawy	2100614
3.	Nabil Ali Ebrahim Fouda	2101592
4.	Abdelrahman Essam Niazy	2101502

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Bill of materials

component	Classification	function	material	Image
Wheel	mechanical	Movement of robot	Carbon-Fiber composite	
Chassis	mechanical	Body of robot	Acrylic	
Arduino UNO	Control& info-processing unit	Control the robot	//	
DC Motor 3-6V	Actuator (electromechanical)	Drive robots' wheels	//	
H-Bridge driver (L298N)	Actuator (electronic)	Control the direction of a DC motor	//	
Line Tracker Module (5 Channels) TCRT5000L	Sensor	Sense nearby obstacles	//	

Color sensor (TCS3200)	sensor	determine the colour of the target object	//	
Battery Holder (3.7v)				
Battery (3.7v 5500mAh)	electrical	Power source for the driver	//	

1- Arduino UNO

[arduino uno datasheet](#)

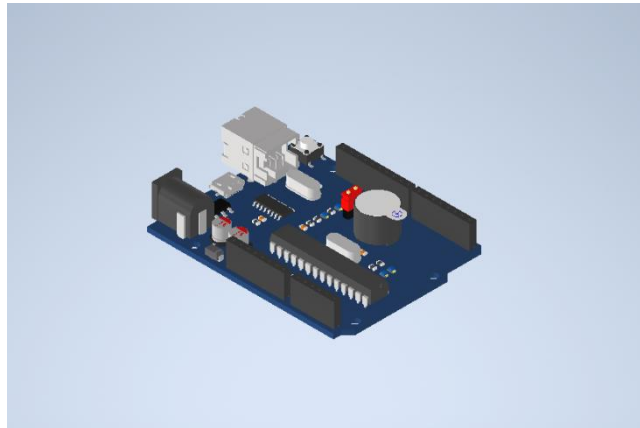


Figure 1: Arduino UNO CAD drawing

2- Motor DC 3-6V

[dc motor data sheet](#)

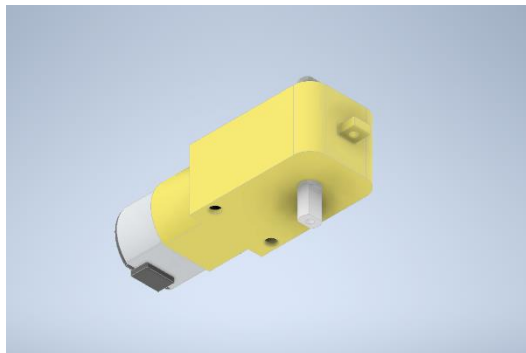


Figure 2: DC motor CAD drawing

3-Line Tracker (TCRT5000L)

[line tracker data sheet](#)

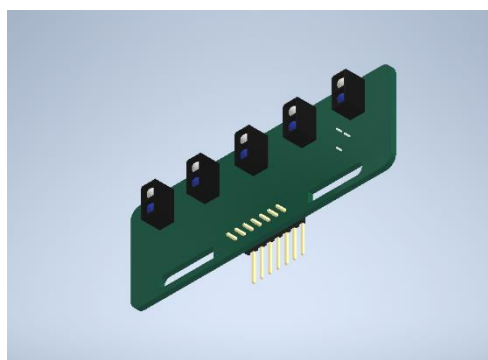


Figure 3: Line tracker CAD drawing

4- H-Bridge driver (L298N)

[h bridge data sheet](#)

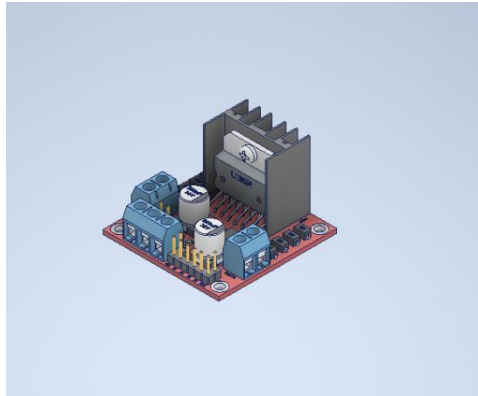


Figure 4: H-bridge CAD drawing

5- Color sensor (TCS3200)

[color sensor data sheet](#)

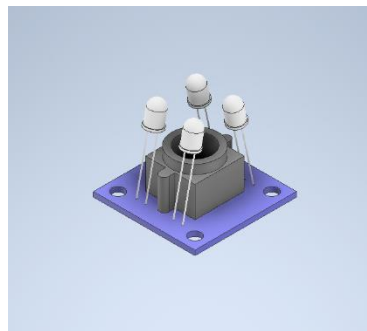


Figure 5: colour sensor CAD drawing

6- Chassis

- Black Acrylic.
- Manufactured with laser cutting.

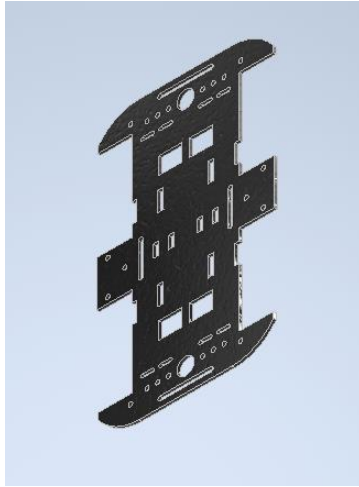


Figure 6: Chassis CAD drawing

7- Wheel D65x25:



Figure 7: Wheel CAD drawing

8-Battery Holder (3.7v)

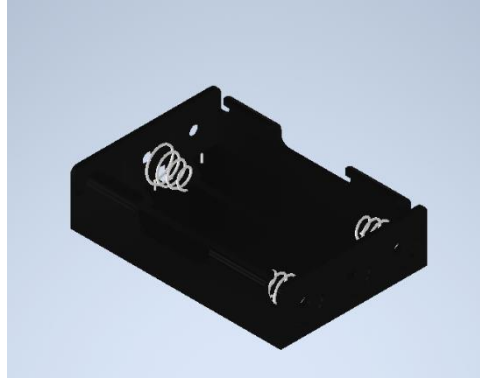


Figure 8: Battery holder CAD drawing

9- Rechargeable Li-ion Battery18650 (3.7v 5500mAh)

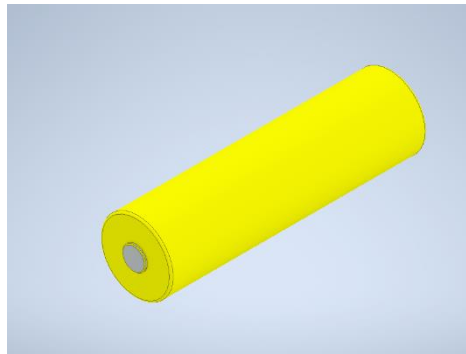


Figure 9: Battery CAD drawing

Drawings

CAD working drawings.

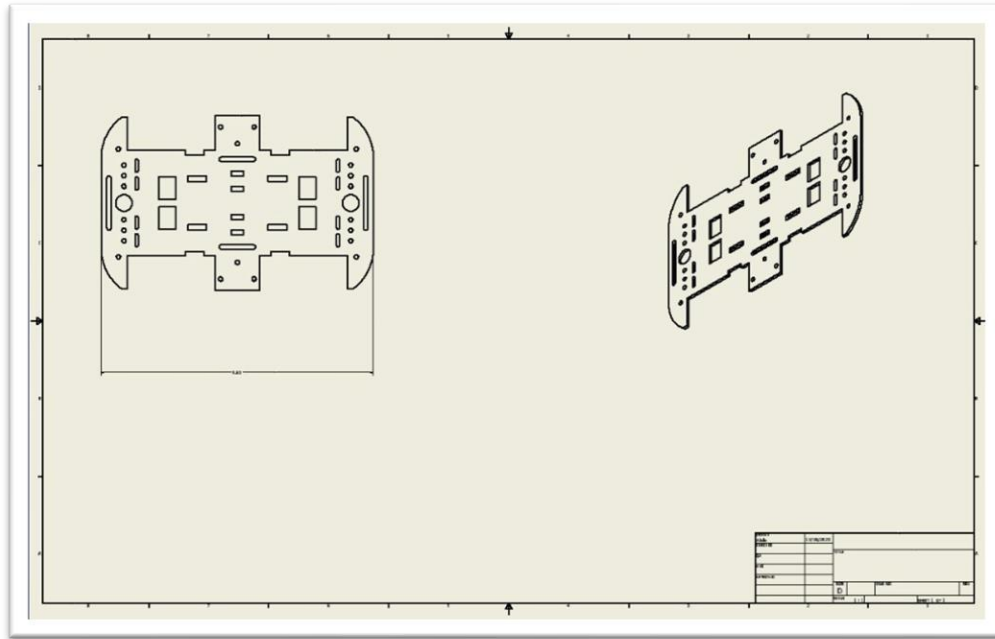


Figure 10: chassis cad working drawing.

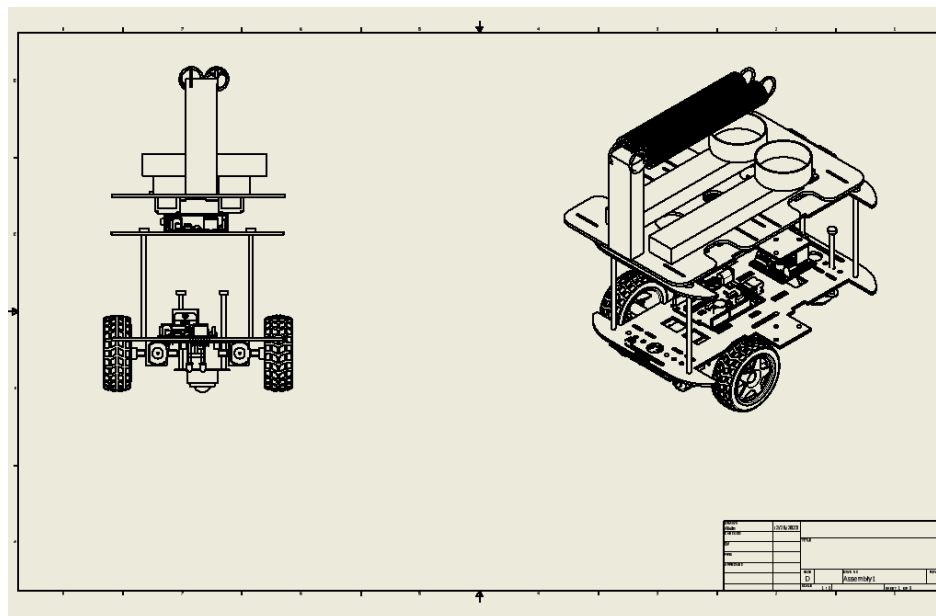


Figure 11: car CAD working drawing.

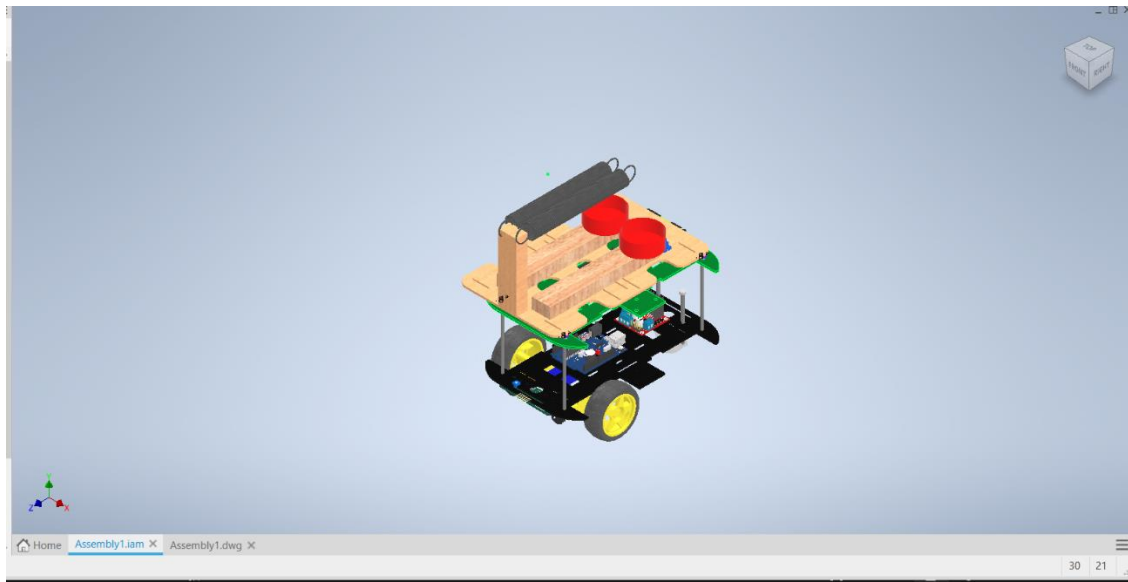


Figure 12: car cad drawing

Stress analysis

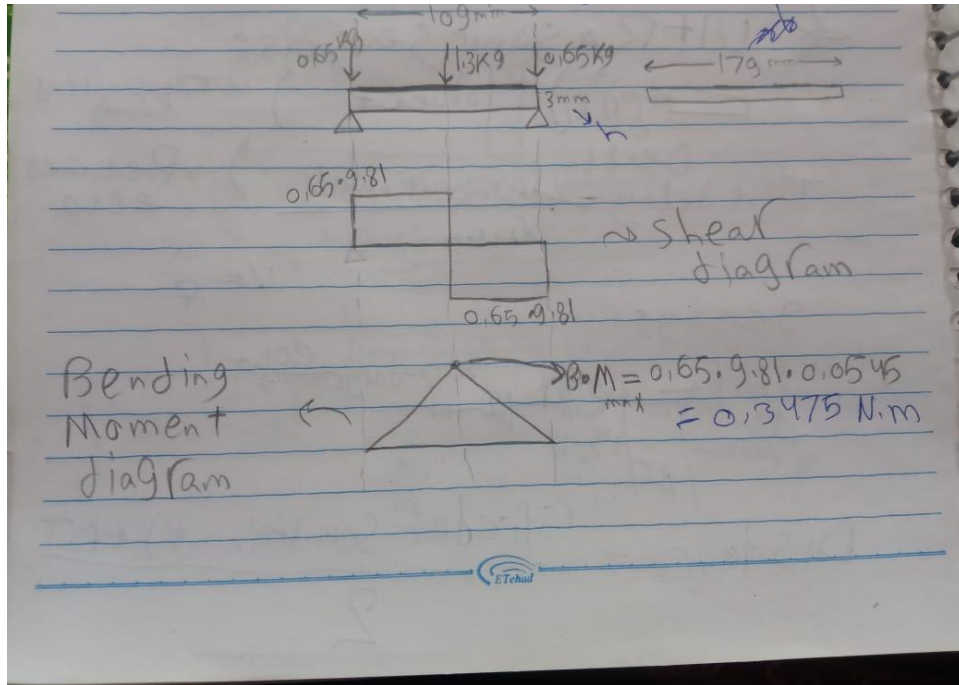


Figure 13: shear diagram & Bending moment Diagram.

Chassis stress analysis

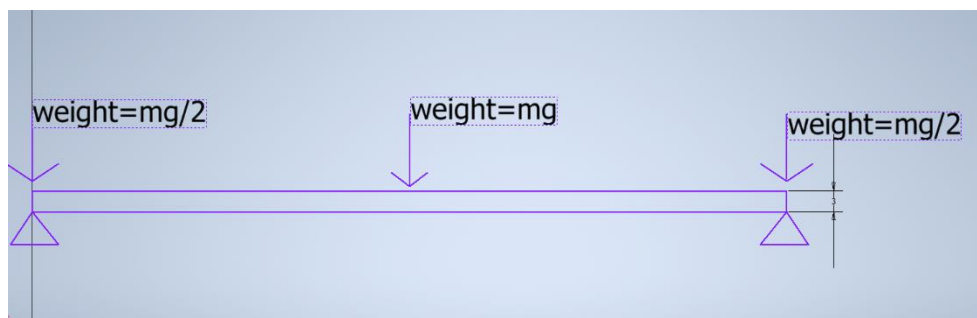


Figure 14: forces acting on the body

Calculations

- $\sigma_{max} = \frac{M_{max} \cdot y}{I} = 2.1255 \text{ MPa}$

$$M_{max}(\text{max bending moment}) = \frac{mgl}{4}$$

Y (distance from the neutral plane) is approximately at the center = 1.5 mm

$$I \text{ (area moment of inertia)} = \frac{b \times h^3}{12}, b = 144\text{mm} \text{ \& } h = 3\text{mm}$$

- Yield stress of chassis material(acrylic) = 50 MPa > 2.1255MPa, then loading on the chassis is safe& won't cause failure.

Motor shaft analysis

- $\tau = \frac{M_{motor} \times r}{J} = 0.02 \text{ MPa}$

r (radius of shaft)

$$J \text{ (polar moment of inertia)} = \pi \frac{r^4}{2}$$

M_{motor} : torque provided by motor

car moves at a constant speed,

$$\sum \text{Torque} = I\ddot{\theta}, \ddot{\theta} = 0$$

$$\text{torque}_{motor} = \text{torque}_{wheel}$$

$$\text{torque}_{wheel} = r_{wheel} \times (\text{weight} + \text{friction force})$$

$$\text{friction force} = \text{normal reaction} \times \mu_k, \mu_k = 1.2$$

$$\text{torque}_{motor} = \frac{mg}{2} r (1 + \mu_k) = 0.24 \text{ N.m}$$

- Max. torsional stress is slightly bigger than the elastic limit of acrylic(which is at 2.76 MPa).
2.76MPa > 0.02,
so the shaft won't fail.

Electrical and Electronic System

PCB layouts

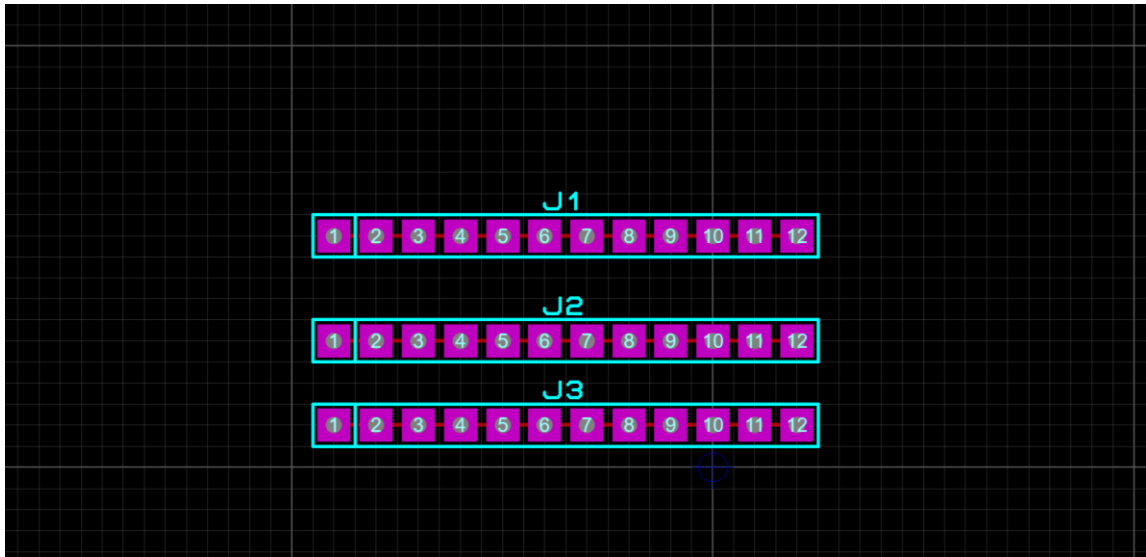


Figure 15: connectors pcb layout

Battery sizing

We tried different voltages to run the motors. We found the suitable voltage to run the motors in the car is 11.1 volts $[11 < V < 12]$ V. we used three rechargeable LI-ION batteries 3.7 volts each in series with a total voltage of 11.1 volts.



Figure 16: LI-ION Batteries

System interconnection diagram

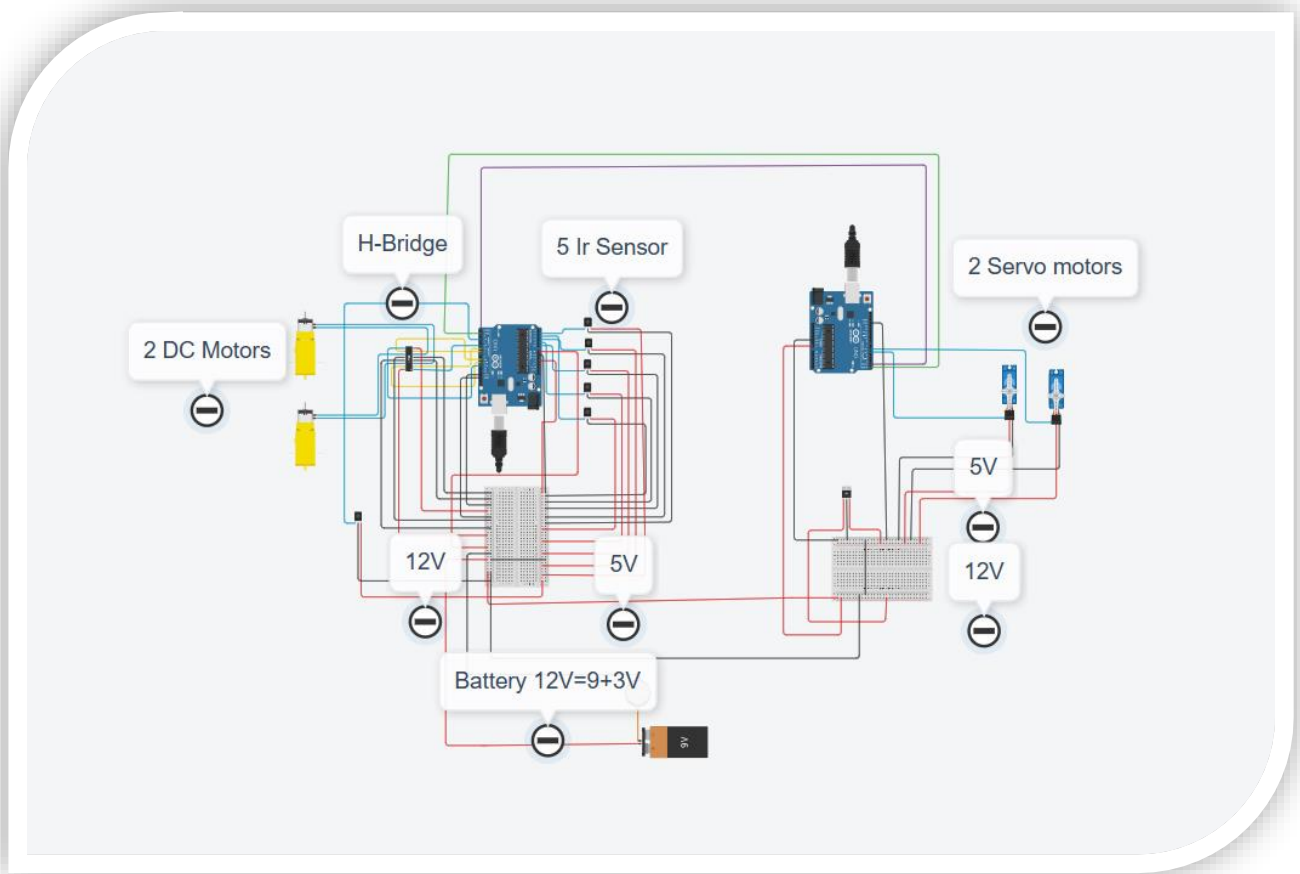
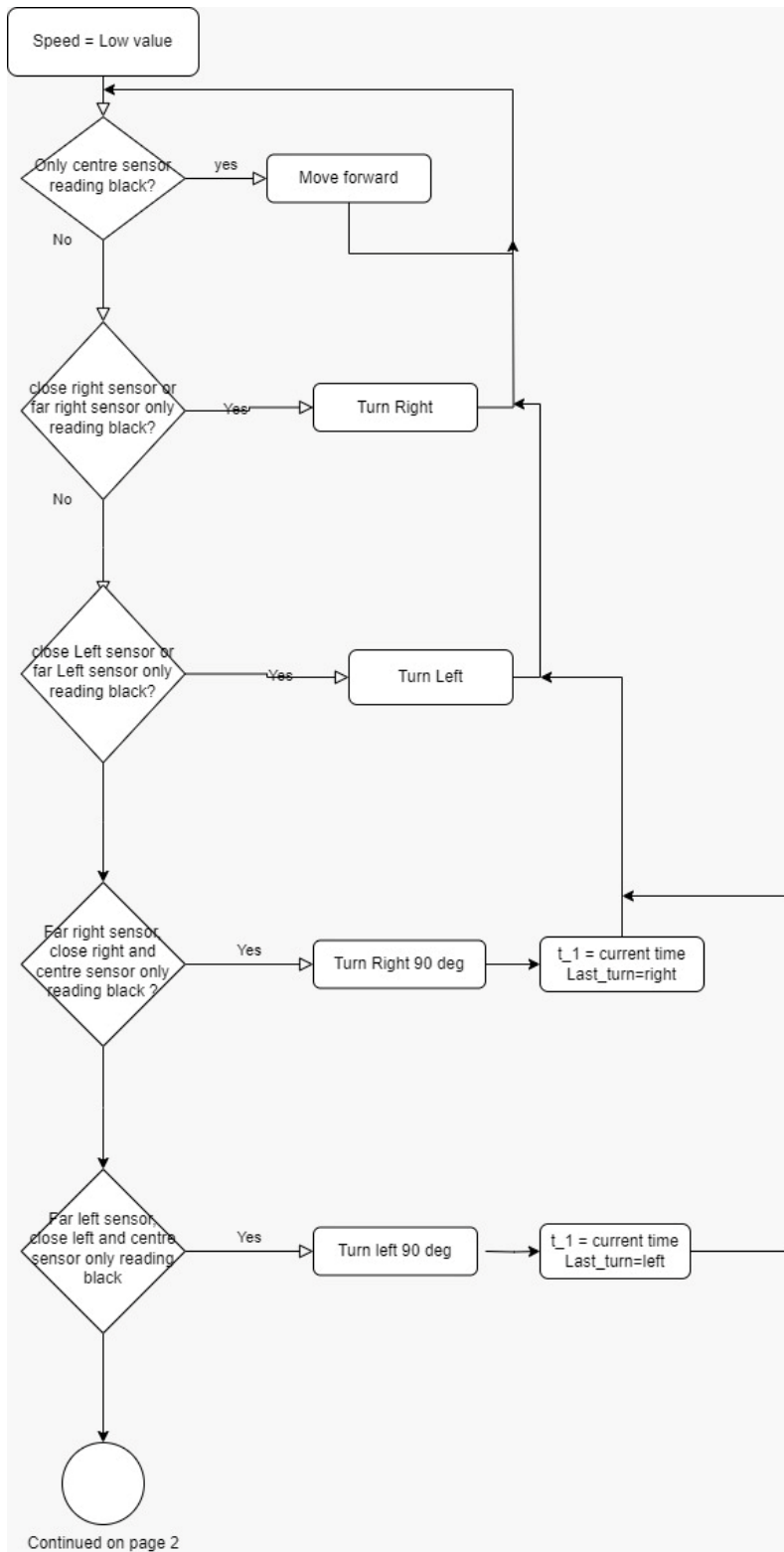


Figure 17: schematic diagram of the project

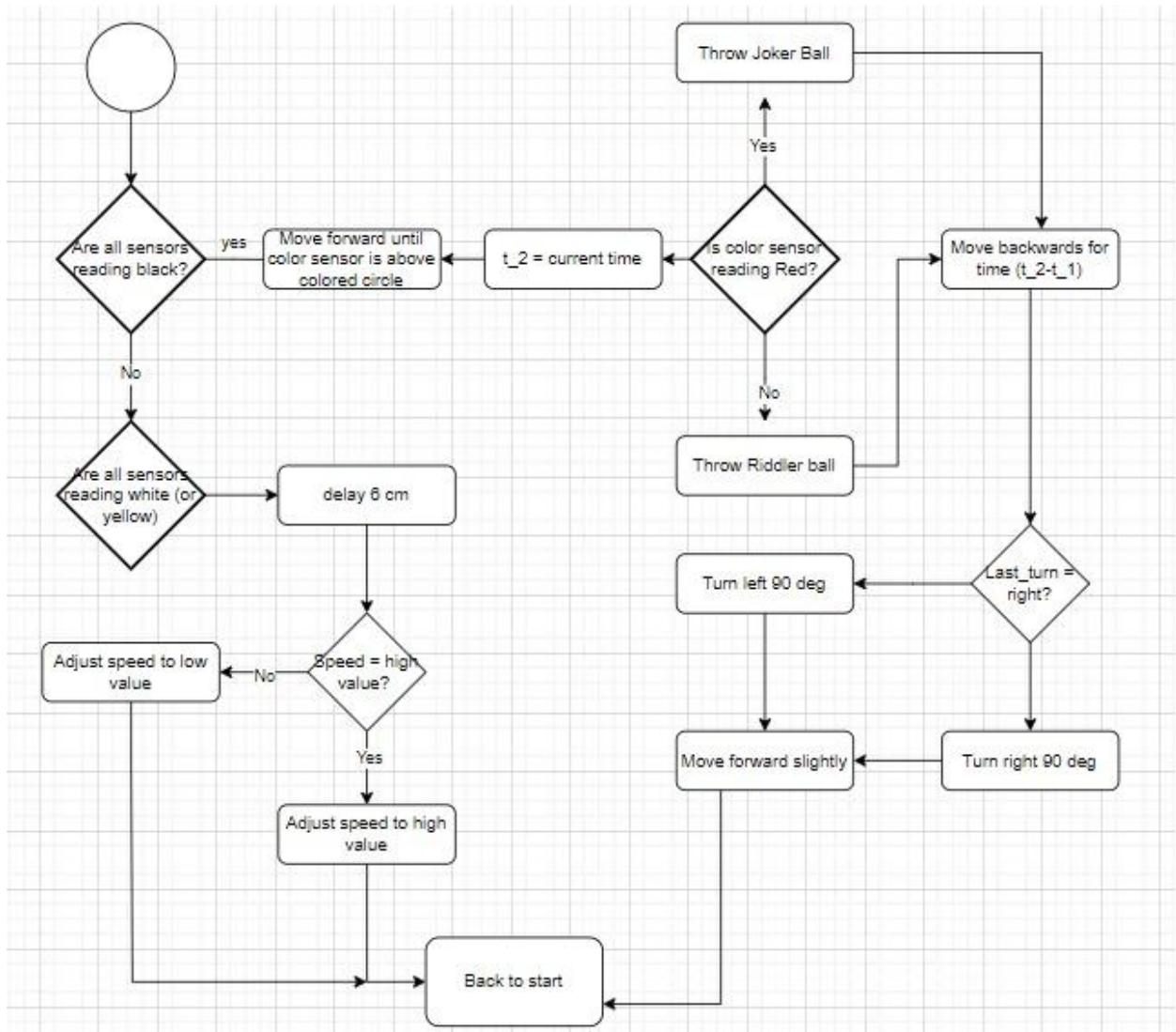
Control System

Flow charts for all control algorithms.

1.



2.



Code Documentation.

```
#define enA 5 // Enable1 L293 Pin enA

#define in1 6 // Motor1 L293 Pin in1

#define in2 7 // Motor1 L293 Pin in2

#define in3 9 // Motor2 L293 Pin in3

#define in4 10 // Motor2 L293 Pin in4

#define enB 11 // Enable2 L293 Pin enB

#define R_S 2 // IR sensor Right

#define L_S A1 // IR sensor Left

#define F_LS A3

#define F_RS A4

#define C_S A0

#define S0 3

#define S1 4

#define S2 11

#define S3 12

#define OUT 13

char msg;

int R, G, B;

int RC=0, GC=0, BC=0, YC=0;


char turn = 'R';

int throw_count = 0;

int R;

int G;
```

```

unsigned long t1;
unsigned long t2;

void setup() {
pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);
  pinMode(OUT, INPUT);
  // Setting frequency-scaling to 20%
  digitalWrite(S0, LOW);
  digitalWrite(S1, HIGH);
  Serial.begin(9600);

  pinMode(R_S, INPUT);
  pinMode(L_S, INPUT);
  Serial.begin(9600);
  pinMode(F_RS, INPUT);
  pinMode(F_LS, INPUT);
  pinMode(enA, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);
  pinMode(enB, OUTPUT);
  analogWrite(enA, 110);
  analogWrite(enB, 110);
}

```

```

void loop() {

  if (digitalRead(R_S) == 1 && digitalRead(L_S) == 1 && digitalRead(C_S) == 0 && digitalRead(F_LS) == 1
  && digitalRead(F_RS) == 1) {

    forward();

    Serial.println("moving forward");

  } else if ((digitalRead(R_S) == 0 && digitalRead(L_S) == 1 && digitalRead(F_RS) == 1 &&
digitalRead(F_LS) == 1 && digitalRead(C_S) == 1) || (digitalRead(R_S) != 0 && digitalRead(L_S) != 0 &&
digitalRead(F_RS) != 1 && digitalRead(F_LS) != 0)) {

    turnRight();

    Serial.println("moving Right");

  } else if ((digitalRead(R_S) == 0 && digitalRead(L_S) == 1 && digitalRead(F_RS) == 0 &&
digitalRead(F_LS) == 1 && digitalRead(C_S) == 0) || (digitalRead(R_S) != 0 && digitalRead(L_S) != 0 &&
digitalRead(F_RS) != 1 && digitalRead(F_LS) != 0)) {

    turnRight();

    Serial.println("moving 90 Right");

    //certain delay

    t1=millis();

    turn ='R';

  } else if ((digitalRead(R_S) == 1 && digitalRead(L_S) == 0 && digitalRead(F_RS) == 1 &&
digitalRead(F_LS) == 1 && digitalRead(C_S) == 1) || (digitalRead(R_S) != 0 && digitalRead(L_S) != 0 &&
digitalRead(F_RS) != 1 && digitalRead(F_LS) != 0)) {

    turnLeft();

    Serial.println("moving Left");

  } else if ((digitalRead(R_S) == 1 && digitalRead(L_S) == 0 && digitalRead(F_RS) == 1 &&
digitalRead(F_LS) == 0 && digitalRead(C_S) == 0) || (digitalRead(R_S) != 0 && digitalRead(L_S) != 0 &&
digitalRead(F_RS) != 1 && digitalRead(F_LS) != 0)) {

    turnLeft();

    Serial.println("moving 90 Left");

    //certain delay

    t1=millis();

    turn ='L';
  }
}

```

```

    } else if ((digitalRead(R_S) == 0 && digitalRead(L_S) == 0 && digitalRead(F_RS) == 0 &&
digitalRead(F_LS) == 0 && digitalRead(C_S) == 0)) {
    Stop();
    delay(100);
    forward();
    delay(300);
    t2= millis()-100;
    Stop();
    Serial.println("T");
    Serial.println("Stop");
    delay(1300);
    backward();
    delay(t2-t1);
    if(turn=='R'){
    turnRight();
    //certain delay
    }
    else{
    turnRight();
    //certain delay
    }

} /*else if (digitalRead(R_S) == 0 && digitalRead(L_S) == 0 && digitalRead(C_S) == 0) {
    backward();
    Serial.println("backward");
} */
if (Serial.available()){
    msg=Serial.read();
    if (msg=='T'){

```

```

    int RC=0, GC=0, ;
    for (int i =0; i<5;i++){// Setting red filtered photodiodes to be read Red frequency
    digitalWrite(S2, LOW);
    digitalWrite(S3, LOW);
    R = pulseIn(OUT, LOW); // Reading the output Red frequency
    delay(100);
    // Setting Green filtered photodiodes to be read Green frequency
    digitalWrite(S2, HIGH);
    digitalWrite(S3, HIGH);
    G = pulseIn(OUT, LOW); // Reading the output Green frequency
    delay(100);
    if (R>G){
        GC+=1;
    }
    else if(G>R){
        RC+=1;
    }

    if(RC>GC){
        Throw_red();
    }
    if(GC>RC){
        Throw_green();
    }

    }

}

```

```
void forward() { // Forward
    digitalWrite(in1, LOW); // Right Motor forward Pin
    digitalWrite(in2, HIGH); // Right Motor backward Pin
    digitalWrite(in4, HIGH); // Left Motor backward Pin
    digitalWrite(in3, LOW); // Left Motor forward Pin
}
```

```
void backward() { // Backward
    digitalWrite(in1, HIGH); // Right Motor forward Pin
    digitalWrite(in2, LOW); // Right Motor backward Pin
    digitalWrite(in4, LOW); // Left Motor backward Pin
    digitalWrite(in3, HIGH); // Left Motor forward Pin
}
```

```
void sharpRight() {
    digitalWrite(in1, LOW); // Right Motor forward Pin
    digitalWrite(in2, HIGH); // Right Motor backward Pin
    digitalWrite(in4, LOW); // Left Motor backward Pin
    digitalWrite(in3, HIGH);
}
```

```
void sharpLeft() {
    digitalWrite(in1, HIGH); // Right Motor forward Pin
    digitalWrite(in2, LOW); // Right Motor backward Pin
    digitalWrite(in3, LOW); // Left Motor backward Pin
    digitalWrite(in4, HIGH);
}
```

```

void Stop() { // Stop

    digitalWrite(in1, LOW); // Right Motor forward Pin
    digitalWrite(in2, LOW); // Right Motor backward Pin
    digitalWrite(in3, LOW); // Left Motor backward Pin
    digitalWrite(in4, LOW); // Left Motor forward Pin
}

```

```

void turnRight() {

    digitalWrite(in1, LOW); // Right Motor forward Pin
    digitalWrite(in2, HIGH); // Right Motor backward Pin
    digitalWrite(in4, LOW); // Left Motor backward Pin
    digitalWrite(in3, LOW);
}

```

```

void turnLeft() { // Turn Left

    digitalWrite(in1, LOW); // Right Motor forward Pin
    digitalWrite(in2, LOW); // Right Motor backward Pin
    digitalWrite(in3, LOW); // Left Motor backward Pin
    digitalWrite(in4, HIGH); // Left Motor forward Pin
}

```

Function prototypes

- void forward () // Forward
- void backward () // Backward
- void sharpRight()
- void sharpLeft()
- void Stop() // Stop
- void turnRight()
- void turnLeft() // Turn Left

developer name per function

Nabil Ali fouda

- void forward ()
- void backward ()
- void sharpRight()
- void sharpLeft()
- void Stop ()
- void turnRight()
- void turnLeft()

System Integration & Performance evaluation

Mechatronic System Configuration Block Diagram.

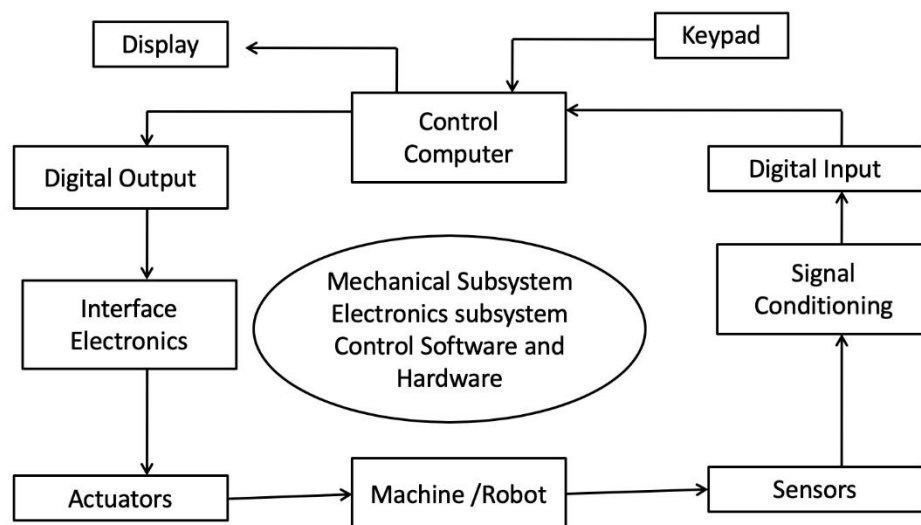


Figure 18: mechatronic system block diagram

Function Structural Diagram.



Figure 19:function structural diagram

All domain specific calculations

Actuator Sizing.

Average(minimum-operating) speed $\approx 90 \pm 10\%$ rpm @ (3-4.5)VDC

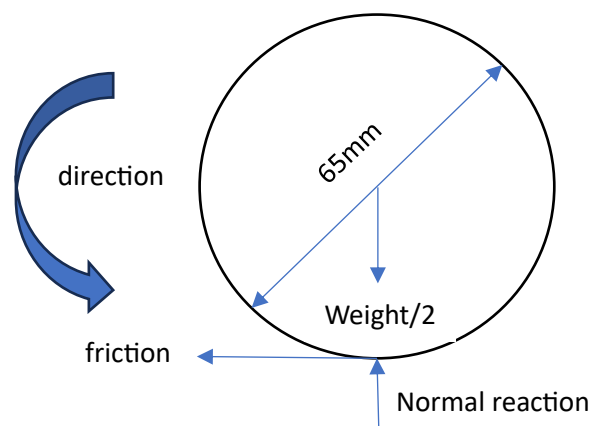


Figure 20: forces on front wheels

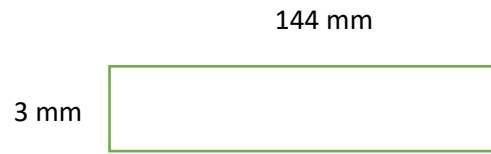


Figure 21: perpendicular cross-sectional area

$$\Sigma Torque = I\ddot{\theta}, \ddot{\theta} = 0$$

$$torque_{motor} = torque_{wheel}$$

$$torque_{wheel} = r_{wheel} \times (weight + friction\ force)$$

$$friction\ force = normal\ reaction \times \mu_k, \mu_k = 1.2$$

$$torque_{motor} = \frac{mg}{2} r(1 + \mu_k) = 0.24 N.m$$

$$\begin{aligned} p_{motor} &= torque_{motor} \times speed_{motor} = 2.26 \text{ watts (at minimum-operating speed of 90 rpm @3VDC)} \\ &= 5.03 \text{ watts (at maximum speed of 90 rpm)} \end{aligned}$$

Timing synchronization and turning movement calibration.

It is measured that the car Arrives at first spot in 24 seconds, Arrives the other spot in 45 secs and completes the whole track in 1 min 10 sec.

[videos link](#)